

Maxime Lafleur

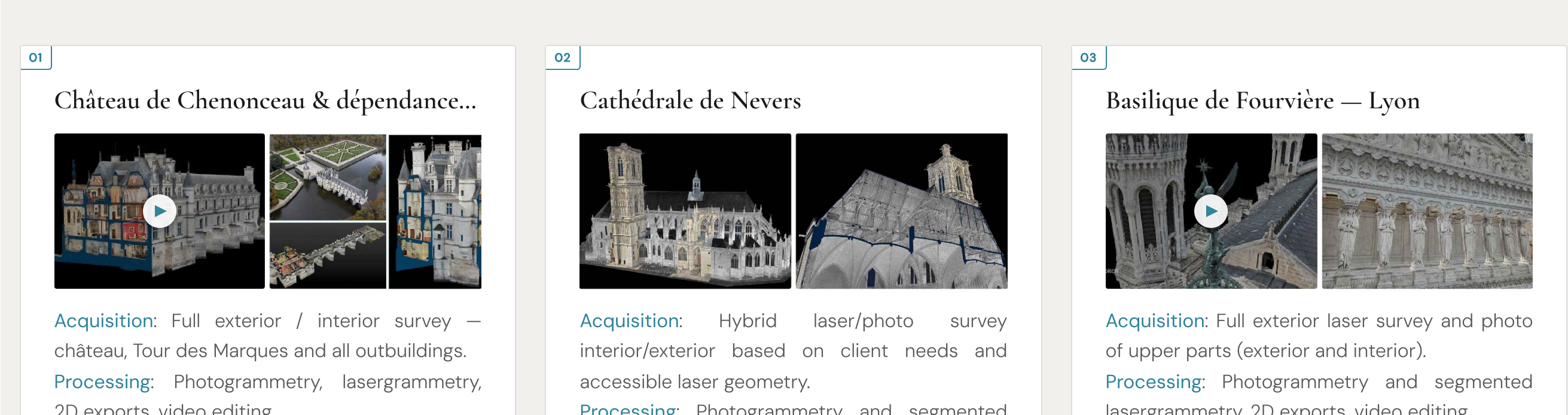
Engineer - Photogrammetrist - Lasergrammetrist - Developer

Passionate about cultural heritage and cutting-edge 3D reconstruction technologies, I initially built my expertise around drone geomatics — from hardware and software development to applied research — gaining full autonomy over the entire process, from study and prototyping to product development (2014–2022).

More recently (2022–2026), my work in the cultural heritage sector immersed me in high-fidelity 3D capture and reconstruction. The technical challenges and the blending of diverse technologies meant mastering specialized skills and methodologies, pushing existing solutions to their limits, and developing automation to eliminate repetitive tasks.

With 3D technology evolving rapidly and opening up new possibilities, I'm eager to keep exploring this field, learn from others and draw on new approaches, as part of a creative team — bringing my 12 years of curiosity and varied experience.

Looking forward to connecting and discussing projects, opportunities and experiences!



Key Heritage & Civil Engineering projects at Geokali

01

Château de Chenonceau & dépendance...

Acquisition: Full exterior / interior survey — château, Tour des Marques and all outbuildings.

Processing: Photogrammetry and lasergrammetry merged into a single 3D model, 2D exports, video editing.

Acquisition: 3 mois / 2 yrs
Images: ~285 000
3D Models: ~150 - 193M Δ
Project size: 16 To

Processing: 9 mois
Scans: ~5 500
Texture resolution: 1mm
2D AutoCAD drawings: 173 .dwg ~700 ortho

03

Cathédrale de Nevers

Acquisition: Hybrid laser/photo survey interior/exterior based on client needs and accessible laser geometry.

Processing: Photogrammetry and segmented lasergrammetry, 2D exports and video editing.

Acquisition: 1 mois
Images: ~40 000
3D Models: ~30 - 237M Δ
2D AutoCAD drawings: 57 .dwg ~200 ortho

Processing: 2 mois
Scans: ~1 830
Texture resolution: 1mm

03

Basilique de Fourvière — Lyon

Acquisition: Full exterior laser survey and photo of upper parts (exterior and interior).

Processing: Photogrammetry and segmented lasergrammetry, 2D exports, video editing.

Acquisition: 2 semaines
Images: ~48 000
3D Models: ~20 - 199M Δ
Project size: 57 .dwg ~200 ortho

Processing: 1 mois
Scans: ~400
Texture resolution: 1mm

04

Église Notre-Dame — Dijon

Acquisition: complete survey of all interior/exterior parts of the church.

Processing: Photogrammetry and lasergrammetry merged into a single 3D model, 2D exports with variants, video editing.

Acquisition: 3 semaines
Images: ~54 000
3D Models: ~2 - 140M Δ
2D AutoCAD drawings: 40 .dwg ~160 ortho

Processing: 2 mois
Scans: ~1 100
Texture resolution: 1mm

05

Église Saint-Philibert — Dijon

Acquisition: full interior and exterior photogrammetric survey excluding roof space.

Processing: Photogrammetry and lasergrammetry merged into a single 3D model, 2D exports, video editing.

Acquisition: 2 semaines
Images: ~30 000
3D Models: ~40 - 198M Δ
2D AutoCAD drawings: 4 .dwg ~40 ortho

Processing: 1 mois
Scans: ~250
Texture resolution: 1mm

06

Civil Engineering Structure

Acquisition: Full structural survey of 3 turbine groups before and after dismantling.

Processing: in-situ and 3D markings, before/after comparisons, 3D and 2D exports.

Acquisition: 2+2 semaines
Images: ~60 000
3D Models: 3 - 5CM Δ
Project size: 8 To

Processing: 9 mois
Scans: ~400
Texture resolution: 0.35 mm
Defects (Optima 315): ~900 m

Key Heritage R&D projects at Geokali

01

Mobile LiDAR data integrated into the photogrammetric pipeline

Integration of mobile scanner / LiDAR data into the standard photogrammetric workflow. Interior with mobile scanner, exterior with mobile scanner supported by photogrammetric capture.

Benefits:

- Maintain a standard 3D production workflow only, replacing static scans with virtual positions from SLAM
- Panoramic photo contribution from the SLAM scanner, equivalent to static panoramas

Costs:

- Georeferencing precision, drift analysis, strict GCP setup — strong mastery of SLAM solutions and their limits
- Risk of geometric fidelity failure (drift, double surfaces) with no correction capacity during processing

Study site: Abbaye de Cluny
Ground photo acquisition: 2 hours
Texture resolution (x ground photogrammetry resolution): 1mm

SLAM acquisition: 2+45 min along trajectory
Scans: 1/2 m

02

Automated 2D DWG sheet generation

Ortho-images + cross-section polylines — automated generation of DWG sheets (plans / sections / elevations).

Benefits:

- Automate DWG extraction generation (plans / sections / elevations)
- Provide an internal operator with an export selection tool for resolutions and client-subscribed options (depth views, contours, confidence map, pure 3D view)
- Eliminate any human error in the processing chain
- Avoid chained operations during extractions
- Systematize deliverables and lock nomenclature, extraction themes and renders
- Respond quickly to client requests — extraction productivity multiplied by x30 per .dwg sheet

Costs:

- SDK software development, trial and error with trusted partners, maintenance across new software versions.

03

Technology watch

Study and benchmark of capture solutions alternative or complementary to static scanners: mobile scanners, mobile SLAM, high-resolution photogrammetry — to optimize acquisition time and adapt methods to site geometry.

04

3D Production by segmentation

Benefits:

- Reducing 3D processing time and complexity
- Splitting work between 3D operators and computers
- Ability to reprocess a specific area following potential changes
- Adapting detail level and texture quality per zone
- Offering clients optimized acquisition costs

Costs:

- Careful cleaning of all laser scans, filtering of images / laser stations, 3D model cleaning, strict file naming conventions.

05

Hardware prototyping

Optimizing acquisition systems by adding purpose-built accessories adapted to field constraints.

Identified needs:

- Accessing confined spaces
- Securing the system for specific use cases
- Simplifying daily operations (heavy handheld equipment)

Adopted solutions:

- Laser scanner trolley (productivity gain)
- 3D camera gimbal for repetitive facades or confined spaces (DJI Ronin)
- Steadycam vest and belt for heavy equipment (Flycam Galaxy)

Custom solutions:

- Low-profile tripod for terrestrial scanner (daily use)
- Camera add-on fixed to terrestrial laser scanner
- Custom attachment systems for gimbals

06

Software prototyping

Pushing daily-use software to its limits to improve working comfort, deliverable quality and processing time, and to develop key algorithms for acquisition / processing methodologies.

Automated script development for:

- AutoCAD / Photoshop / Leica Cyclone 3DR / Agisoft Metashape / RealityScan

Developed tools:

- Multi-scale analysis tool for alignment discontinuities from RealityScan photogrammetric processing
- Automated image QC tool (motion/focal blur, exposure) with renaming, copying and raw data backup
- Ortho-image generation with various renders: depth, contours, Xray, 3D confidence, geometric voids — to streamline the drafter's workflow

Small algorithms:

- Image filtering by location and orientation
- WGS84 to local projected coordinates conversion
- Storage redundancy detection script
- Automated conversion of panoramic to equirectangular images
- Injection of pre-calibrated camera calibrations into EXIF metadata
- And other small useful developments on a project-by-project basis

PROFESSIONAL EXPERIENCE

Geokali
2022 → JUNE 2026 - 4 YRS

Microdrones™ / GeoCue
2016 → 2022 - 6 YRS 5 MO

CHRS
2021 → 2022 - 1 YR 6 MO

Delair
2016 → 2018 - 2 YRS

INVENTUAV
2015 → 2016 - 1 YR

Vision du Ciel Intl
2014 → 2016 - 1 YR

Groupe SII - Nexter
2013 → 2014 - 1 YR

Photogrammetry / Lasergrammetry Engineer

Geokali - Full-time - Valence, France

Oct. 2022 → present

3D data acquisition and processing on **cultural heritage** and **civil engineering** sites, using a full sensor fleet: UAV drones, terrestrial laser scanners (TLS), mobile laser scanners (MLS), DSC cameras and structured light lasers.

→ [See Heritage & Civil Engineering projects](#) · [See R&D projects](#)

Jean-Marc Anneron — Draftsman-Surveyor-Laser scanning specialist, Historic monuments since 2001

“ I had the pleasure of working alongside Maxime Lafleur for several years on demanding operations related to surveying and graphical restitution of historic monuments. Looking back (and as a Heritage draftsman with over 25 years of experience), I can say without hesitation that he is among the best specialists in the field I have had the chance to collaborate with. Maxime has an extremely rare profile, combining technical rigidity, adaptability, creativity and rigor. Where many see a blocker or a constraint, he already starts designing a solution he codes, develops routines, automates complex tasks and creates efficient methodologies that push 3D/2D tool outputs to their maximum. The quality, readability and ease of use of his deliverables reflect this. In the field, his mastery of capture operations is remarkable. Whether in 3D digitization (photogrammetry, laser scanning), he shows great rigor and an excellent reading of situations, which perfectly matches and feeds Geokali's expertise. But beyond the skills, what I probably appreciate most about him is the human solidity he brings to a collaboration. Maxime is approachable, reliable, attentive, proactive and generously involved. When working with him, you immediately feel you can move forward with confidence, even on the most complex projects.”

PHOTOGRAMMETRY | LIDAR-TLS/MLS | UAV | 3D RECONSTRUCTION | LEICA SDK | DWG AUTOMATION | POINT CLOUDS | HARDWARE PROTOTYPING

CULTURAL HERITAGE | R&D

UAV LiDAR Engineer

Microdrones™ / GeoCue - R&D - Toulouse

May 2016 → Sept. 2022

Within the R&D department of Microdrones / GeoCue, responsible for developing **new UAV-embedded LiDAR prototypes**, designing high-precision LiDAR targets, and QC of new processing software.

- Operational setup of Microdrones France subsidiary (office design, construction follow-up, HR)
- Hardware design of UAV LiDAR systems
- Field testing and data QC
- Hardware and software benchmarks
- Development of an automated tree counting tool
- Automated powerline classification
- Parachute system development and EASA certification approval

Vincent Eucatat — Industrial CAD Draftsman at midGroup

“ I had the opportunity to work alongside Maxime Lafleur during my time at Microdrones, and I can wholeheartedly vouch for his excellent skills as an engineer. As part of our drone data acquisition missions for various vineyard estates in France, Maxime was a true technical pillar. His analytical mindset and great resourcefulness allowed him to efficiently diagnose and solve complex hardware and software issues, often under demanding field conditions. Whether it was last-minute adjustments to the equipment or optimizing flight operations, his ingenuity always ensured the continuity and quality of our operations. I highly recommend him to any organization looking for a reliable and talented field engineer.”

Erwan Gavelle — Founder of geopid

“ I worked with Maxime at Delair on several tests and projects related to drones and imaging. Friendly and particularly rigorous, he enjoys working as part of a team. We also share a particular interest in several areas outside of work, which only strengthened our collaboration.”

UAV LIDAR R&D | HARDWARE DESIGN | EASA CERTIFICATION | POWERLINE CLASSIFICATION | TREE COUNTING | FIELD TESTING

3D / GIS Support for Antarctic PhD Project

CNRS - Civic Service - Remote

Feb. 2021 → Jun. 2022

Technical and methodological support for a **PhD project** on the **spatial distribution of penguins in Antarctica**. 3D processing of 11 Antarctic sites.

- GNSS base station PPP post-processing
- PPK LiDAR processing
- Multi-site photogrammetry
- High-precision Ortho-DEM generation

ANTARCTICA | PPK LiDAR | GNSS PPP | PHOTOGRAMMETRY | ORTHO-DEM | GIS

UAV Engineer Payload Integration & Control/Command

Delair - Full-time - Toulouse

May 2016 → May 2018

Within Delair-Tech's R&D department, under Microdrones' strategic direction, responsible for:

- Control/command system development for Microdrones UAVs
- Integration of Delair-Tech payloads (multispectral, optronic, photogrammetry) on Microdrones md4-1000
- Viticulture research project in partnership with Naïo Technologies (vineyard row mapping with PPK)
- Other prototyping projects

Christophe Eucatat — Head of Operations

“ I had the chance to work with Maxime on the development of drone payloads. He was in charge of the electronics and project management while I worked on the mechanics. It was my first job in this field, but he trusted me, taught me and challenged me throughout the projects. I particularly appreciated his skills, his involvement and his friendliness in driving the projects to success. He also managed a large part of the organization and setup of our offices in France. I highly recommend Maxime for his professionalism, his good mood and his ability to support the people he works with.”

Pierre-Louis Tiliket — Director at Tialcio

“ I had the chance to work for several years with Maxime Lafleur when he was in charge of integrating Microdrones products into the Delair ground station, as well as developing and integrating several payloads, including multi-spectral sensors. He is capable of managing application software development, embedded development, as well as handling flight tests, operations, and processing of photogrammetric data, showing a thorough rigor and a complete skill set. He is an extremely motivated, reliable and passionate person who, once he takes part in a project, won't let go until it works. I strongly recommend him, as with over 10 years of experience he will be a solid ally for your projects.”

PAYLOAD INTEGRATION | FIXED-WING UAV | EMBEDDED SYSTEMS

Co-Founder

InventUAV - Open-Source Startup - France

May 2015 → May 2016

Co-founder of an open-source UAV startup developing an **autonomous indoor drone** for warehouse management and inventory.

- ArduPilot plugin for Gazebo / ROS
- ArduCopter debugging
- Visual recognition for indoor missions

ARDUPILOT | ROS / GAZEBO | OPEN SOURCE | INDOOR DRONE

UAV Apps Engineer (GCS / GUI)

Vision du Ciel Intl - Orange

Jul. 2014 → Jun. 2015

Development of **UAV mission control** software solutions on Windows: mission planning, control, supervision and reporting.

- GUI development in C#, JS, Python
- Embedded development in C++
- Open-source and industrial protocol integration
- Prototyping: UAVs, radio controllers, gimbals, parachutes

GCS DEVELOPMENT | C# / PYTHON / JS | PIXHAWK | HARDWARE PROTOTYPING

System Architect Nexter Systems

Groupe SII - Satory

Oct. 2013 → Jul. 2014

Writing, analysis and capitalisation of system requirements for Nexter. Architecture in IBM DOORS, document traceability, automation script development, object-oriented C++ modelling.

SYSTEM ARCHITECTURE | IBM DOORS | C++ | REQUIREMENTS ANALYSIS

More information available on request

TECHNICAL EXPERTISE

Skills & tools (non exhaustive)

Acquisition & Field

- Hardware prototyping (tripod, steadycam, gimbal)
- UAV / drone (multirotor, fixed-wing)
- Camera calibration (intrinsic & extrinsic)
- Aerial & terrestrial photogrammetry
- MLS — Mobile Laser Scanning and Lidars
- Field quality control (QA/QC)
- GNSS / PPK / PPP processing
- TLS — Terrestrial Laser Scanning
- RDV underwater drones

Software

- RealityCapture / RealityScan / PostShot / Metashape
- Point cloud processing & cleaning
- GIS (QGIS)
- 2+F Laser Control Office / Scantia
- Unreal Engine 5 / Cinema 4D
- GNSS processing (BPK/PPP)
- Solidworks & 3D printing
- Leica Cyclone3DR
- Postshot & 3DGS
- CloudCompare
- Claude AI
- AutoCAD
- Blender

Development

- Multi-source 3D reconstruction (photo + TLS + MLS + structured light)
- Automated tools (plan extraction, QC, AI denoising)
- AI experiments (photogrammetry, Android apps)
- ROS / Gazebo
- System architecture & requirements (LMI, DOORS)
- ArduPilot / APM
- Processing pipeline (farm processing)
- Scanner hardware benchmarking
- GUI development (Windows, GCS mission control)
- Electronic prototyping
- DWG export / automated 2D drawings
- Payload integration
- Python
- EXIF scripting
- C++ / C#

Photography passion

Film photography & scanning, night photography, astrophotography

CURIOSITY | PATIENCE | RIGOR

Azores Project: discovering the world of sailing

Buying the sailboat, learning sailing & navigation principles, onboard medical training, test trips, equipment/safety audit, and the north-rip voyage.

SELF-TEACHING | RIGOROUS PREPARATION | SELFLESSNESS | MENTAL / PHYSICAL RESILIENCE | FREEDOM | ACCEPTANCE OF DISCOMFORT

Archiving / passing on knowledge

Photogrammetry of stereoscopic family tree research and photo archiving historical research

CURIOSITY | PATIENCE | RIGOR | JOY OF SHARING KNOWLEDGE

Support

Managing a non-profit website, opening a CAF mountain hut, sailboat delivery, mechanical help, 3D design & printing, Android apps

JOY OF TEACHING | PATIENCE | SELF-TEACHING | RIGOR

Highlands Project: cycling

Scotland Highlands project: route mapping, wild camping and self-sufficiency

PREPARATION | ORGANIZATION & MENTAL MANAGEMENT | FREEDOM

Desert Discovery

Planning and carrying out self-sufficient routes in Iceland / Morocco / Tunisia / Mauritania then later guiding friends and family.

MAPING | PREPARATION AND PATIENCE | ORGANIZATION | TEAM MANAGEMENT | FLEXIBILITY & ATTENTIVENESS TO OTHERS | SENSE OF FREEDOM | ACCEPTANCE OF DISCOMFORT

Active pursuits

Trekking, mountaineering, competition climbing

COMMITMENT | DEDICATION | RISK ACCEPTANCE | MENTAL MANAGEMENT | SAFETY | SENSE OF FREEDOM

Aviation passion

Paragliding training and practice, discovering the physics of gliders/planes/helicopters, simulators

CURIOSITY | MENTAL MANAGEMENT | RIGOR | PATIENCE

RESEARCH & INNOVATION

Publications & Patents

01 2020

Patent : 3D Accuracy Star + Catenary Lines

PATENT

02 2020

Validation of LiDAR Survey Data by Comparison of Several Uncertainty Models

PUBLICATION

03 2020

Comparison of UAV LiDAR Odometry of Rotating and Fixed Velodyne Platforms

PUBLICATION

04 2019

Denoising of 3D Point Clouds

PUBLICATION

05 2017

Tri-Rotors UAV Stabilization for Vertical Takeoff and Hovering

PUBLICATION

EDUCATION

Education & International exchanges

EU

European Drone Certificate

Open AI/A2/A3 and Specific CAFE

2026

EE

ECE Paris

Master of Engineering (MEng) Electrical & Electronics Engineering

2008–2013

US

University of California, Irvine

Exchange Program Global Management & Manufacturing

2012

CA

Concordia University

Semester Exchange Engineering & Computer Science

2011

CONTACT

Let's connect !

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MOBILITY: Rhône-Alpes → France → Europe → World

LANGUAGES: French (native) - English (Full Professional)